

TAZEWELL COUNTY AP, VA, USA

WMO: 720264

Lat: 37.064N

Lon: 81.798W

Elev: 808

StdP: 91.98

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-11.0	-23.9	0.5	2.2	-19.6	0.7	-2.0	8.7	2.3	7.9	2.4	2.9	300	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	30.2	21.7	28.8	21.1	27.7	20.6	23.8	27.7	22.9	26.6	22.2	25.8	2.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	19.0	25.8	22.1	18.5	25.3	21.2	17.5	24.2	75.7	27.9	72.4	26.9	69.6	25.2	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.5	5.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-17.9	24.4	2.0	1.4	-19.4	25.4	-20.6	26.2	-21.7	27.0	-23.2	28.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	0.8	2.9	6.3	12.5	17.0	20.9	22.4	22.2	19.5	13.5	7.8	3.0		
	DBStd	9.08	6.76	6.37	5.80	5.51	4.27	2.61	2.05	2.22	3.33	4.94	5.03	6.02		
	HDD10.0	1039	290	209	144	38	4	0	0	0	0	25	99	229		
	HDD18.3	2599	542	433	373	184	77	8	1	2	26	161	318	474		
	CDD10.0	1929	6	11	30	112	221	326	383	377	286	132	32	13		
	CDD18.3	445	0	1	0	8	34	84	126	120	61	10	1	0		
	CDH23.3	2468	0	0	0	36	170	472	712	682	351	45	1	0		
(13)	CDH26.7	512	0	0	0	2	17	98	160	159	71	6	0	0		
(14)	Wind	WSAvg	2.2	2.9	3.0	3.0	2.0	1.7	1.5	1.3	1.6	2.1	2.4	2.5		
(15) Precipitation	PrecAvg	1117	88	83	111	92	115	108	128	102	75	63	71	87		
	PrecMax	1426	159	200	186	175	212	183	251	170	150	131	154	184		
	PrecMin	861	39	26	42	38	52	51	57	46	22	12	16	34		
	PrecStd	170	33	37	42	36	48	39	40	32	29	31	34	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.3	20.6	22.3	26.3	28.5	31.2	32.1	32.0	30.9	27.2	21.2	18.7		
		MCWB	13.5	15.2	12.3	15.5	19.5	21.5	23.2	23.0	19.4	17.3	12.4	13.1		
	2%	DB	14.8	17.0	19.0	24.0	26.9	28.9	29.9	29.9	28.6	24.1	18.7	16.2		
		MCWB	11.7	12.2	10.3	14.1	18.4	20.6	22.3	22.2	19.4	16.2	9.8	12.0		
	5%	DB	12.6	14.0	17.3	22.5	25.4	27.6	28.2	28.3	27.3	22.4	17.2	14.1		
		MCWB	9.6	10.4	10.5	13.7	17.4	20.2	21.6	21.7	18.9	15.1	11.1	11.1		
	10%	DB	11.0	12.2	15.4	20.9	23.8	26.3	27.3	27.3	25.3	20.8	15.3	12.4		
		MCWB	8.4	8.5	8.9	13.4	16.8	19.5	21.4	21.1	18.5	14.8	8.7	9.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	15.2	14.6	17.4	21.3	23.8	24.8	24.9	22.9	20.0	15.8	14.8		
		MCDB	16.1	19.2	19.1	22.8	26.4	27.6	29.3	28.7	26.5	24.2	18.9	16.9		
	2%	WB	12.3	13.7	13.2	16.1	20.0	22.6	24.0	23.7	21.4	18.8	14.0	13.5		
		MCDB	14.2	16.2	16.8	21.4	24.8	26.8	28.0	27.7	25.3	22.2	16.3	15.2		
	5%	WB	10.6	11.6	12.1	15.2	19.1	21.8	23.0	22.8	20.6	17.4	12.6	11.5		
		MCDB	12.2	13.7	15.5	20.3	23.6	25.9	26.5	26.5	24.6	20.1	15.1	13.3		
	10%	WB	8.5	9.1	10.2	14.1	18.1	21.0	22.3	21.9	19.7	16.1	11.1	9.8		
		MCDB	9.9	11.2	14.2	19.0	22.1	24.8	25.6	25.6	23.8	19.1	13.7	12.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.9	9.3	10.2	9.8	9.6	9.2	9.1	9.5	9.2	9.1	7.5		
		MCDBR	9.3	9.7	11.0	11.0	10.8	11.1	10.8	10.7	11.5	10.9	10.2	7.7		
		MCWBR	7.1	7.0	7.1	5.7	5.1	5.2	4.6	4.6	4.6	5.4	6.5	5.9		
	5% WB	MCDBR	8.7	8.6	9.9	9.5	10.1	9.8	9.4	9.2	9.6	8.7	8.8	8.0		
		MCWBR	7.7	7.3	7.5	5.8	5.3	5.0	4.7	4.5	4.5	5.3	7.1	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.295	0.317	0.356	0.393	0.421	0.449	0.454	0.358	0.331	0.299	0.282		
	taud		2.536	2.503	2.485	2.386	2.276	2.228	2.184	2.143	2.407	2.491	2.555	2.572		
	Ebn at Noon		916	947	951	925	890	861	833	820	893	891	892	899		
	Edh at Noon		77	90	101	118	134	140	146	149	108	90	75	70		
(44) All-Sky Solar Radiation	RadAvg		2.11	2.73	3.88	5.00	5.54	6.04	5.62	5.27	4.49	3.45	2.55	1.86		
	RadStd		0.19	0.35	0.29	0.38	0.43	0.39	0.36	0.26	0.41	0.43	0.27	0.19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (35 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+141	+82	

Nomenclature: See separate page